

HW

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Heat equation: $u_t = v u_{xx} + f(x, t)$, $v > 0$.

Energy: $E(t) = \int_0^1 \frac{1}{2} u^2 dx$. We consider $E'(t) = \int_0^1 u u_t dx$

$$\text{so } E'(t) = \int_0^1 u (v u_{xx} + f(x, t)) dx = v \int_0^1 u u_{xx} dx + \int_0^1 u f dx$$

$$\because \int_0^1 u u_{xx} dx = u u_x \Big|_0^1 - \int_0^1 u_x^2 dx = u(1, t) u_x(1, t) - u(0, t) u_x(0, t) - \int_0^1 u_x^2 dx$$

$$\therefore E'(t) = v [u(1, t) u_x(1, t) - u(0, t) u_x(0, t)] - v \int_0^1 u_x^2 dx + \int_0^1 u f dx$$

By $u(x, t) = 1$ is the exact sol. of $u_t = v u_{xx} + f(x, t)$, so for $u \equiv 1 \Rightarrow u_t = 0, u_{xx} = 0$

$$\therefore u(0, t) = u(1, t) = 0, v(u u_x) \Big|_0^1 = 0 \Rightarrow E'(t) = -v \int_0^1 u_x^2 dx \leq 0 \text{ (when } f=0)$$

but $u \equiv 1$ does not satisfy Dirichlet BCs, $u=0$ at endpoints, $\therefore$ hence, the Dirichlet BCs:

$$E'(t) = -v \int_0^1 u_x^2 dx \leq -v C E(t) \Rightarrow E(t) \leq E(0) e^{-2v C t} \text{ is failing (} u \equiv 1 \neq 0)$$

↑
by Poincaré inequality.

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